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AN OUTBREAK OF HEPATO-ENTERITIS (THE PALM ISLAND MYSTERY DISEASE) POSSIBLY CAUSED BY ALGAL INTOXICATION

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INTRODUCTION

During November 1979 an outbreak of hepato-enteritis occurred on Palm Island which is located north-east of Townsville in the Australian State of Queensland. Clinical details have already been described (BYTH, 1980). Hepato-enteritis in its severe form had three well-defined phases which presented in the following order: the hepatitis phase of two days' duration; the lethargic phase resulting from severe electrolyte derangement; and the diarrhoeal phase usually of five days' duration which commenced one to two days after the lethargic phase. It is the purpose of this presentation to summarise the findings of the epidemiological investigation and to give reasons for postulating that this outbreak was due to blue-green algal intoxication.

MATERIALS AND METHODS

Clinical and laboratory data were obtained from the staff of the Townsville General Hospital, the Palm Island Hospital and the Commonwealth Pathology Laboratory in Townsville. Additional clinical and environmental specimens were submitted to the Laboratory of Microbiology and Pathology and the Government Chemical Laboratory.

An environmental inspection was conducted to determine the sanitary status of Palm Island. A survey of households was undertaken and questionnaires dealing with the consumption of green mangoes, fruit-bats and water, were distributed to 100 school children.

BACKGROUND

During October 1979 an algal bloom occurred in Solomon Dam, the sole source of the reticulated water supply on Palm Island. The bloom discoloured the water and gave it a disagreeable odour and taste. To remedy the situation experiments were undertaken to obtain untainted water by drawing it from various depths of the dam's intake well. The outcome of these efforts were successful but short-lived. When the bloom became dense, it was treated with one part per million of copper sulphate between the 12th and 14th of November.

Because of the poor quality of the piped water caused by the bloom, the community serviced by the reticulation system commenced to obtain their water from a number of springs and unprotected shallow wells in, or close to, the residential area. Five days after completion of the copper sulphate treatment, the first case of hepato-enteritis occurred.

Solomon Dam, which had recently been constructed, was only filled to capacity for the first time during the summer rains of late 1978 and early 1979. During that period significant amounts of non-aquatic vegetation were submerged.

RESULTS

The initial clinical impression was that the illness was due to chemical intoxication. Clinical investigations and the laboratory examination of environmental material however failed to identify the aetiological agent of the outbreak.

When the epidemiological investigation was conducted, 136 persons of Aboriginal descent had contracted hepato-enteritis. The age range of those involved was from two to 28 years. Forty-four per cent of the cases occurred in children between the ages of five and 14 years. Cases less than five years of age accounted for 23 per cent of the total. At the termination of the outbreak 139 children and ten adults of Aboriginal descent had contracted the illness (BYTH, 1980).

The characteristics of the epidemic suggested that the outbreak was of a common-source type. The epidemic reached its peak eight days after its commencement and lasted for 21 days. By the thirteenth day of the outbreak over two thirds of the cases had been accounted for. No secondary cases occurred in household contacts. There was no geographical clustering or temporal patterns of spread in the main residential area serviced by the reticulated water supply. But it was observed that no illness had been reported among approximately 50 persons living in an isolated area near the aerodrome. The houses in that locality were not connected to the reticulated water supply. Water was obtained by those residents from a nearby unprotected shallow well.

A survey of affected households gave on the whole little information. It was felt that those questioned were only prepared to give answers that would cause the least embarrassment to themselves and their families. Questionnaires distributed to 100 school children did statistically implicate green mangoes and wells located in the area serviced by reticulated water. However, most of those who had eaten green mangoes had also ingested well water. All the children surveyed had drunk the reticulated water at some unspecified points in time. The older Aboriginal residents were of the opinion that the illness was associated with the consumption of green mangoes.

Repeated sanitary assessments of the environment of Palm Island failed to uncover the presence of any poisons capable of producing hepato-enteritis. No cross-connections were detected between either the reticulated water supply or unprotected shallow wells and the sewerage, septic tanks or waste-water disposal systems. Mouldy or abnormally bitter-tasting foods had not been sold to the population prior to the outbreak. At the time of the survey no such foods were being offered for sale.

On completion of the epidemiological investigation, advice was sought from four tropical disease experts. They suggested that the outbreak might have been caused by a vegetable toxin (Personal communications: DOWDLE, 1980; PROZESKY and GEAR, 1980; WYATT, 1980).

DISCUSSION

This episode had a number of distinctive features. Children were the main victims and the illness did not appear to be communicable. Infants and all but ten adults were spared. The clinical manifestations were those of an acute intoxication having three well-recognised phases. The outbreak was confined geographically not only to one island but also to a settlement serviced by a reticulated water supply.

No illness was reported in a small group of people living near the aerodrome where the residences were not connected to piped water. And, finally, the outbreak occurred shortly after copper sulphate had been used to treat a dense algal bloom in the sole source of the reticulated water supply.

Recently it has been suggested that the illness on Palm Island was an outbreak of fruit-bat toxocariasis (MOORHOUSE, 1980). This is based, in part, on the observation that the skins of ripe mangoes collected by some of the authors on the island during and after the latter half of the outbreak contained embryonated ova of Toxocara pteropodis, a parasite of the fruit bat, Pteropus alecto, and also on the fact that hepatomegaly is a manifestation of canine toxocariasis in man. While T. pteropodis might cause some degree of morbidity among the population on Palm Island, it seems doubtful that it was responsible for a circumscribed, short-lived outbreak of hepato-enteritis. The acute clinical features alone do not match those observed in persons suffering from canine toxocariasis, which, it is assumed for want of information, may resemble fruit-bat toxocariasis in man. Whereas fruit bats and mangoes have a wide distribution in Queensland, the epidemic on the other hand was confined to only one of two settlements on Palm Island. Although the geographical distribution of mangoes bearing the ova of T. pteropodis on their skins is as yet unknown in Queensland, a contaminated sample from Yeppoon has been reported (MOORHOUSE, 1980). This finding, at least, shows that this fruit-bat parasite is not confined to Palm Island.

After receiving the opinions of the overseas experts, the possibility that the treatment of the dense algal bloom with copper sulphate was in some way associated with the outbreak was then seriously considered. It has long been known that some blooms of blue-green algae in dams, shallow fresh-water lakes, and stagnant pools can cause illness and death in animals (McBARRON and MAY, 1966; JACKSON et al., 1982; SCHANTZ, 1970; SCHWIMMER and SCHWIMMER, 1964). What is generally less well known is that they can also cause intoxication in man. Two reviewers have documented a number of human episodes (SCHWIMMER and SCHWIMMER, 1964).

The toxic factors of noxious blue-green algae are endotoxins which are released after the organisms die, or are physically disintegrated (FALCONER et al., 1982; GENTILE and MALONY, 1969; HUGHES et al., 1958; RUNNEGAR et al., 1982; SIRENKO et al., 1976). Massive fish kills have occurred following the treatment of dense algal blooms with copper sulphate (SAWYER et al., 1968). One of these endotoxins has been shown to be extremely stable in water and is not removed by the usual water-treatment methods used in Australia (FALCONER and RUNNEGAR, 1980). A diarrhoea-producing endotoxin has been obtained from standing laboratory cultures of Microcystis aeruginosa (AZIZ, 1974).

Not all strains of known noxious species of blue-green algae are toxic, and both toxic and non-toxic strains can be isolated from the same bloom. Furthermore, the NRC-1 strain of Microcystis aeruginosa has been shown to be genetically heterogeneous for toxin production (GORHAM, 1964). It appears that environmental factors, such as organic nutrients, light and temperature, influence not only the production of blooms but also enhance the toxicity of some noxious blue-green algae (FALCONER and RUNNEGAR, 1980; GORHAM, 1964; RUNNEGAR et al., 1982; VINYARD, 1967).

Experiments in animals have shown that members of certain genera of blue-green algae from natural blooms and algal cultures can give a wide spectrum of clinical manifestations and pathological effects (SCHWIMMER and SCHWIMMER, 1964). Endotoxins of species of Microcystis, Anabaena and Nostoc can produce clinical pictures resembling that of hepato-enteritis in both naturally and experimentally intoxicated

animals. Species belonging to the genera, Microcystis, Anabaena and Aphanizomenon, have been associated with gastrointestinal disorders in man (SCHWIMMER and SCHWIMMER, 1964).

The paucity of reported incidents of human algal intoxication is probably due to man's reluctance to drink discoloured, foul-tasting, malodorous water. Also, there is presently no method for detecting and identifying endotoxins in isolated cases or outbreaks of intoxication. No doubt these episodes are attributed to other causes and are thus missed.

It is therefore postulated that the outbreak of hepato-enteritis on Palm Island was caused by the presence of endotoxins in the reticulated water supply. These toxins were introduced into the piped system after a dense algal bloom in Solomon Dam had been treated with copper sulphate which resulted in the mass death and decay of blue-green algae belonging to one or more of the genera, Microcystis, Anabaena and Nostoc. The submersion of terrestrial vegetation when the dam was first filled to capacity nine months previously probably provided the high levels of organic nutrients which subsequently favoured the dense growth of noxious blue-green algae.

Why children were mainly affected is just not known. They may be more ready to drink tainted water or, possibly, they may be more sensitive to algal endotoxins. Until additional scientific information on the health hazards of algal blooms becomes available, all dams and similar sources of water supply, especially those of recent construction, should be monitored for algal content and promptly treated with a suitable algicide when a bloom is likely to occur.

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